CHANDIGARH UNIVERSITY

PROJECT REPORT

Topic

Development of a Calculator Program Using Functions and Control Structures

Submitted By	Dixit Sharma
UID	26BCD10279
Course	BCA – Data Science
Subject	Programming in C
Project	Default Course Project – Question 1

Academic Session: 2026–2027

1. Introduction

A calculator is a basic and useful computer application that performs mathematical calculations. This project develops a menu-driven calculator in the C programming language. It allows the user to perform addition, subtraction, multiplication and division by selecting an operation from a menu and entering two numbers.

The program is designed using **functions** and **control structures**. Separate functions are used for each arithmetic operation, while a switch statement selects the required operation. A loop allows the user to perform multiple calculations without restarting the program.

2. Objectives of the Project

The main objectives are:

• To understand the fundamentals of the C programming language.
• To use user-defined functions for modular programming.
• To apply decision-making statements such as switch and if.
• To use loops to repeat a set of instructions.
• To perform the four basic arithmetic operations accurately.
• To handle invalid choices and division by zero.

3. Software and Hardware Requirements

Component	Requirement
Programming Language	C
Compiler / IDE	GCC, Code::Blocks, Dev-C++ or any C compiler
Operating System	Windows, Linux or macOS
Hardware	Computer or laptop with a keyboard

4. Concepts Used

4.1 Functions: A function is a named block of code that performs a specific task. This project uses separate functions for addition, subtraction, multiplication and division. It makes the program easier to read, test and maintain.

4.2 Switch Statement: The switch statement checks the user's menu choice and transfers control to the matching case. It is suitable for selecting one operation from several fixed options.

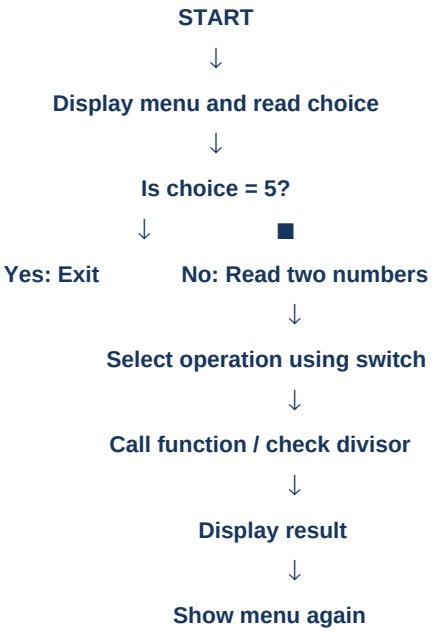
4.3 Do-While Loop: The do-while loop repeats the calculator menu until the user chooses to exit. Since the condition is checked after the loop body, the menu appears at least once.

4.4 If-Else Statement: An if-else condition checks whether the divisor is zero before division. Division by zero is undefined, so the program displays an error message instead of calculating a result.

5. Algorithm

1. Start the program.
2. Display the calculator menu.
3. Read the user's choice.
4. If the choice is 5, exit the program.
5. Otherwise, ask the user to enter two numbers.
6. Use switch to identify the selected operation.
7. Call the corresponding arithmetic function.
8. For division, first check whether the second number is zero.
9. Display the result or an appropriate error message.
10. Repeat the menu until the user selects Exit.
11. Stop the program.

6. Flow of the Program



7. Source Code

The following C program uses four user-defined functions. The main function controls the menu, accepts input and displays the result.

```
#include <stdio.h>

/* Functions for arithmetic operations */
double add(double a, double b) {
    return a + b;
}

double subtract(double a, double b) {
    return a - b;
}

double multiply(double a, double b) {
    return a * b;
}

double divide(double a, double b) {
    return a / b;
}

int main() {
    int choice;
    double num1, num2, result;

    do {
        printf("\n===== SIMPLE CALCULATOR =====\n");
        printf("1. Addition\n");
        printf("2. Subtraction\n");
        printf("3. Multiplication\n");
        printf("4. Division\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice >= 1 && choice <= 4) {
            printf("Enter first number: ");
            scanf("%lf", &num1);
            printf("Enter second number: ");
            scanf("%lf", &num2);
        }

        switch (choice) {
            case 1:
                result = add(num1, num2);
                printf("Result = %.2lf\n", result);
                break;

            case 2:
                result = subtract(num1, num2);
                printf("Result = %.2lf\n", result);
                break;

            case 3:
                result = multiply(num1, num2);
                printf("Result = %.2lf\n", result);
                break;

            case 4:
                if (num2 == 0) {
                    printf("Error: Division by zero is not allowed.\n");
                } else {
                    result = divide(num1, num2);
                    printf("Result = %.2lf\n", result);
                }
                break;

            case 5:
                printf("Calculator closed. Thank you!\n");
                break;

            default:
```

```
                printf("Invalid choice. Please select 1 to 5.\n");
            }
        } while (choice != 5);

        return 0;
    }
```

8. Sample Output

The following is an example of how the calculator works when a user performs different operations:

```
===== SIMPLE CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 1
Enter first number: 25
Enter second number: 15
Result = 40.00

===== SIMPLE CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 3
Enter first number: 8
Enter second number: 5
Result = 40.00

===== SIMPLE CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 4
Enter first number: 20
Enter second number: 0
Error: Division by zero is not allowed.

===== SIMPLE CALCULATOR =====
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 5
Calculator closed. Thank you!
```

9. Explanation of Output

In the first example, the user selects addition and enters 25 and 15. The add() function returns 40. In the second example, multiplication of 8 and 5 gives 40. In the third example, the user attempts to divide by zero; the program detects this condition and displays an error message. Finally, selecting option 5 terminates the program.

10. Testing

Test Case	Input / Choice	Expected Result
Addition	12 + 8	20
Subtraction	18 – 6	12
Multiplication	7 × 4	28
Division	20 ÷ 5	4
Division by zero	20 ÷ 0	Error message

Test Case	Input / Choice	Expected Result
Exit	Choice 5	Program closes

11. Advantages and Limitations

Advantages

• Simple and user-friendly menu.
• Uses separate functions, improving code organization.
• Supports integer and decimal values through double data type.
• Handles division by zero safely.
• Allows repeated calculations in one run.

Limitations

• It performs only the four basic arithmetic operations.
• It is a console-based program and has no graphical interface.
• It does not store calculation history.

12. Conclusion

The Calculator Program project successfully demonstrates how basic arithmetic operations can be implemented in C using functions and control structures. The program performs addition, subtraction, multiplication and division through a simple menu. The switch statement selects the operation, the do-while loop allows repeated use, and the if-else condition prevents division by zero. Through this project, the concepts of modular programming, user input, decision-making and looping are applied in a practical way.

13. Future Scope

The calculator can be improved by adding more mathematical operations such as modulus, percentage, square root and power. A graphical user interface and calculation history can also be added in a future version.

14. References

1. E. Balagurusamy, *Programming in ANSI C*.
2. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*.
3. C standard library documentation for stdio.h and input/output functions.
4. Class notes and practical learning materials on C programming.

— End of Project —